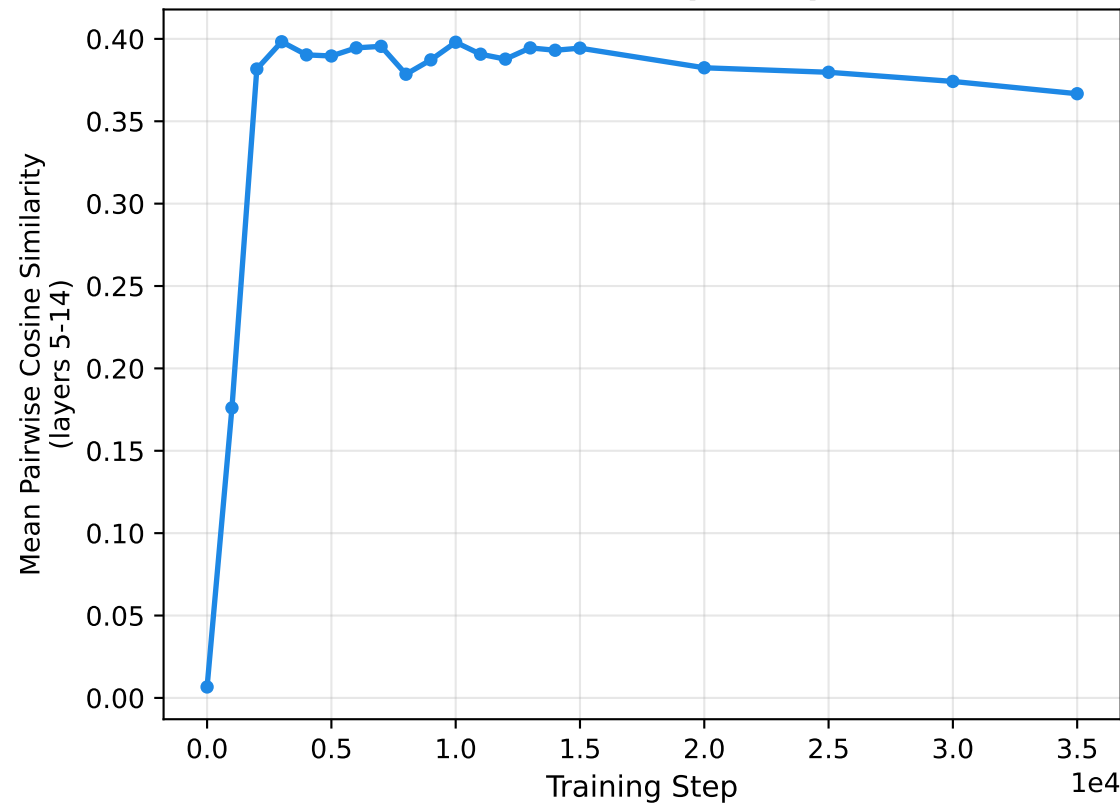
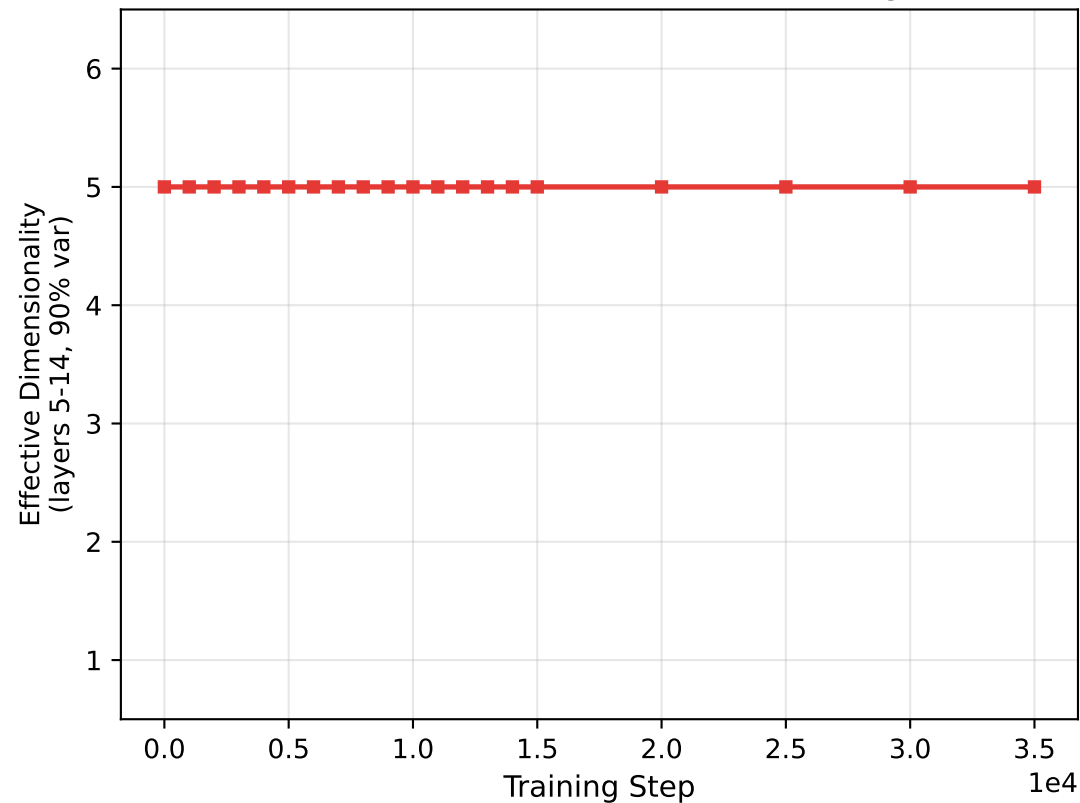


Geometric Trajectory During Pre-Training (OLMo-2 1B)

(a) Framework Collapse/Separation



(b) Framework Direction Diversity



(c) Accuracy vs. Geometry

